

Hoare Logic Practice: October 8, 2026 (20 points)

Consider the following Dafny program that inefficiently calculates four plus the product of two integers. The integer p may be positive, zero, or negative:

```
method ProductPlusFour(n:int, p:int) returns (y:int)
requires n >= 0
ensures y == 4 + n*p
{
  y := 4;
  var x := p;
  var z := 0;
  while (z < n)
  {
    z := z + 1;
    y := y + x;
  }
}
```

Fill in the annotations below to show that the Hoare triple given by the outermost pre- and postconditions is valid. Be completely precise and pedantic in applying Hoare rules: write assertions in exactly the form given by the rules, rather than just logically equivalent ones. Working backward from the end should require the rule of consequence only at the places marked $\rightarrow$. Those implication steps may silently rely on the usual rules of arithmetic.

Edit the nine answer commands near the top of this file. Labels A–I are worksheet labels, not part of the assertions. This is an original practice question modeled on the 2024 midterm. No answer key is included.

- $\{\{ n \geq 0 \}\}$
 $\rightarrow$
- A $\{\{ ((4 = 4 + 0 \cdot p) \wedge (p = p)) \wedge (0 \leq 0 \leq n) \}\}$
 $y := 4;$
- B $\{\{ ((y = 4 + 0 \cdot p) \wedge (p = p)) \wedge (0 \leq 0 \leq n) \}\}$
 $\text{var } x := p;$
- C $\{\{ ((y = 4 + 0 \cdot x) \wedge (x = p)) \wedge (0 \leq 0 \leq n) \}\}$
 $\text{var } z := 0;$
- D $\{\{ ((y = 4 + z \cdot x) \wedge (x = p)) \wedge (0 \leq z \leq n) \}\}$
 $\text{while } (z < n) \{$
- E $\{\{ (((y = 4 + z \cdot x) \wedge (x = p)) \wedge (0 \leq z \leq n)) \wedge (z < n) \}\}$
 $\rightarrow$
- F $\{\{ ((y + x = 4 + (z + 1) \cdot x) \wedge (x = p)) \wedge (0 \leq z + 1 \leq n) \}\}$
 $z := z + 1;$
- G $\{\{ ((y + x = 4 + z \cdot x) \wedge (x = p)) \wedge (0 \leq z \leq n) \}\}$
 $y := y + x;$
- H $\{\{ ((y = 4 + z \cdot x) \wedge (x = p)) \wedge (0 \leq z \leq n) \}\}$
 $\}$
- I $\{\{ (((y = 4 + z \cdot x) \wedge (x = p)) \wedge (0 \leq z \leq n)) \wedge \neg(z < n) \}\}$

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$$\{ \{ y = 4 + n * p \} \}$$